Fullstack Development

Preflight project - backend

Github Repo

Backend applications

- Data storage
- Business logic
- Authentication / authorization
- APIs for frontend frameworks (*depending on architecture*)
- Connection to other services

Node JS backend frameworks

- Github stars
- State of JS 2025

Packages

- `pnpm init`
- Express JS
  - `pnpm i express cors helmet morgan debug`
- Typescript
  - `pnpm i typescript @tsconfig/node-lts @tsconfig/node-ts tsx tsc-alias`
  - `pnpm i -D @types/cors @types/express @types/debug @types/morgan @types/node cross-env nodemon`
- ORM
  - `pnpm i drizzle-orm postgres dotenv`
  - `pnpm i -D drizzle-kit`

Package explanation

Package	Details
`express`	Express *(duh!)*
`helmet`	Set default HTTP response header
`cors`	Enable Cross-origin resource sharing (CORS)
`morgan`	HTTP request logger
`debug`	Debugging utility
`nodemon`	Script monitoring during dev

ORM code

- All files in `./db` folder.
  - You can change schema.
- `./drizzle.config.ts` (Link)

Files

- 💾 `./.env` (Copy from here)
- 💾 `./nodemon.json` (Link)
- 💾 `./.gitignore` (Link)
- 💾 `./tsconfig.json` (Link)
- Scripts in `package.json` (Link)

Minimal example

- 💾 `./src/index.ts` (Link)
- Sync database (if you have not done so)
  - `pnpm run db:push`
- Start dev
  - `pnpm run dev`

Full example

- 💾 `./src/index.ts` (Link)
- Build
  - `pnpm run build`
- Start production
  - `pnpm run start`

API Spec

- Insomnia
- Postman

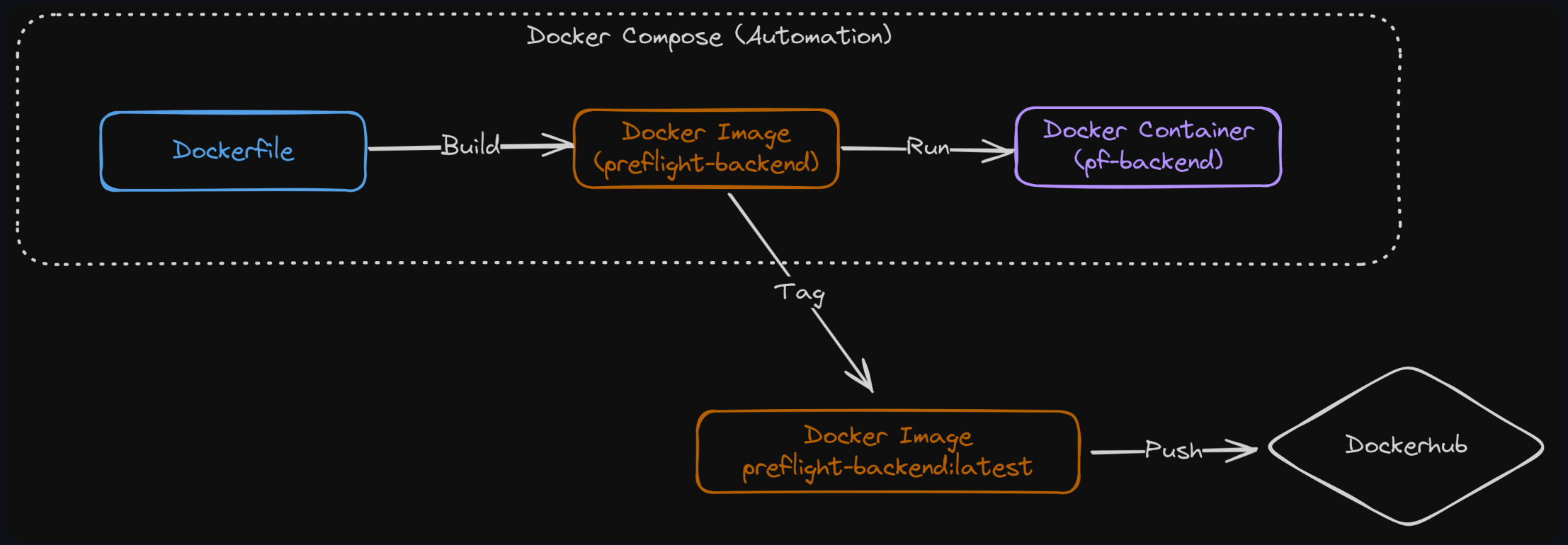
Debugging (optional)

-  `./.vscode/launch.json` (Link)

Migration

- Run before the containerization step.
- `pnpm run db:generate`

Containerization


Docker Compose (Automation)
Dockerfile
Build
Docker Image
(preflight-backend)
Run
Docker Container
(pf-backend)
Tag
Docker Image
preflight-backend:latest
Push
Dockerhub

Steps

- 💾 `./Dockerfile` (Link)
- 💾 `./dockerignore` (Link)
- 💾 `./docker-compose.yml` (Link)
- 💾 `./.env.test` from `./.env.test.example` (Link)
- ⌨️ `docker compose --env-file ./.env.test up -d --force-recreate --build`

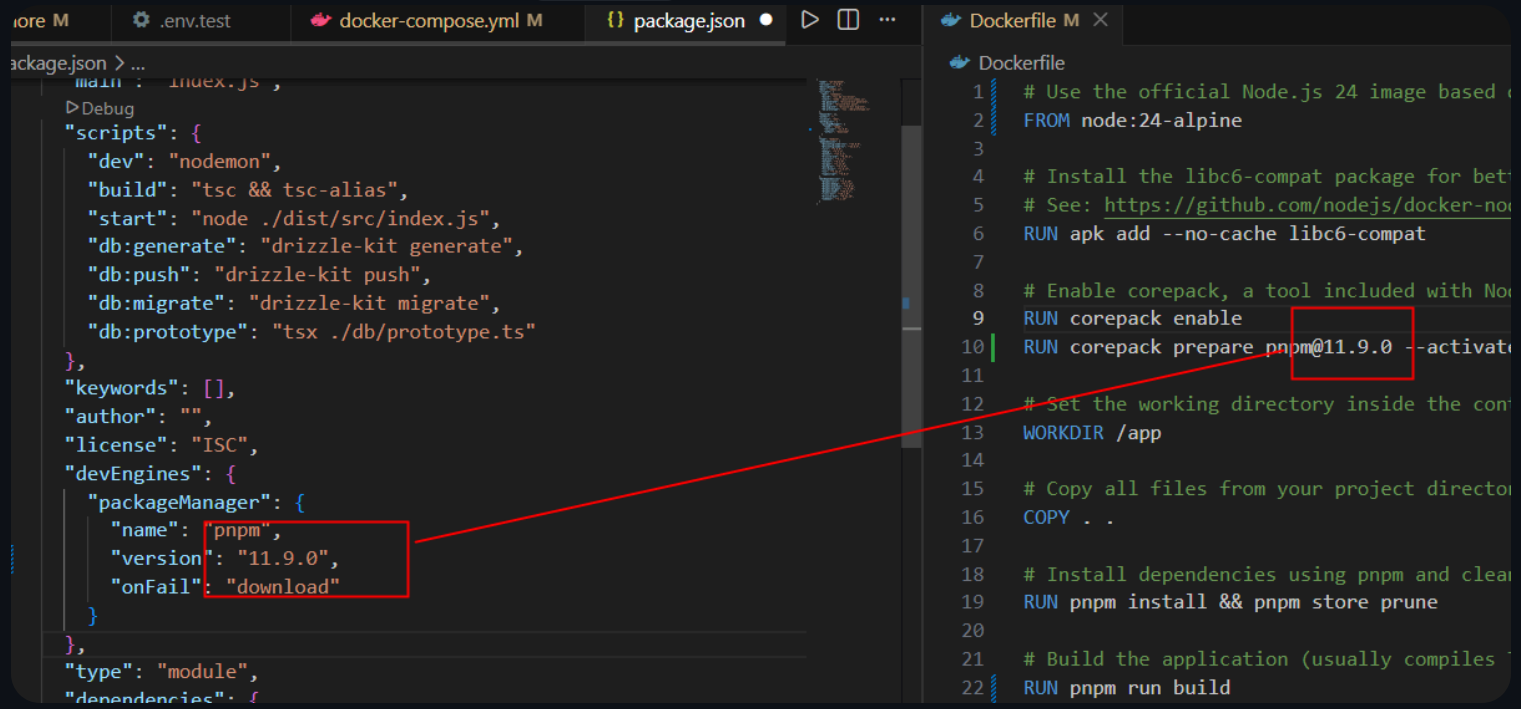
Corepack

- In a `Node-Alpine` Dockerfile, `Corepack` is an official Node.js tool used to install and manage package managers like pnpm and yarn.
- Corepack require an exact SemVer format (e.g., `10.0.0` or pnpm@9.1.0).
  - They will crash if they encounter a range modifier or invalid syntax.

```
  "packageManager": {
    "name": "pnpm",
-   "version": "^11.9.0",
+   "version": "11.9.0",
    "onFail": "download"
```

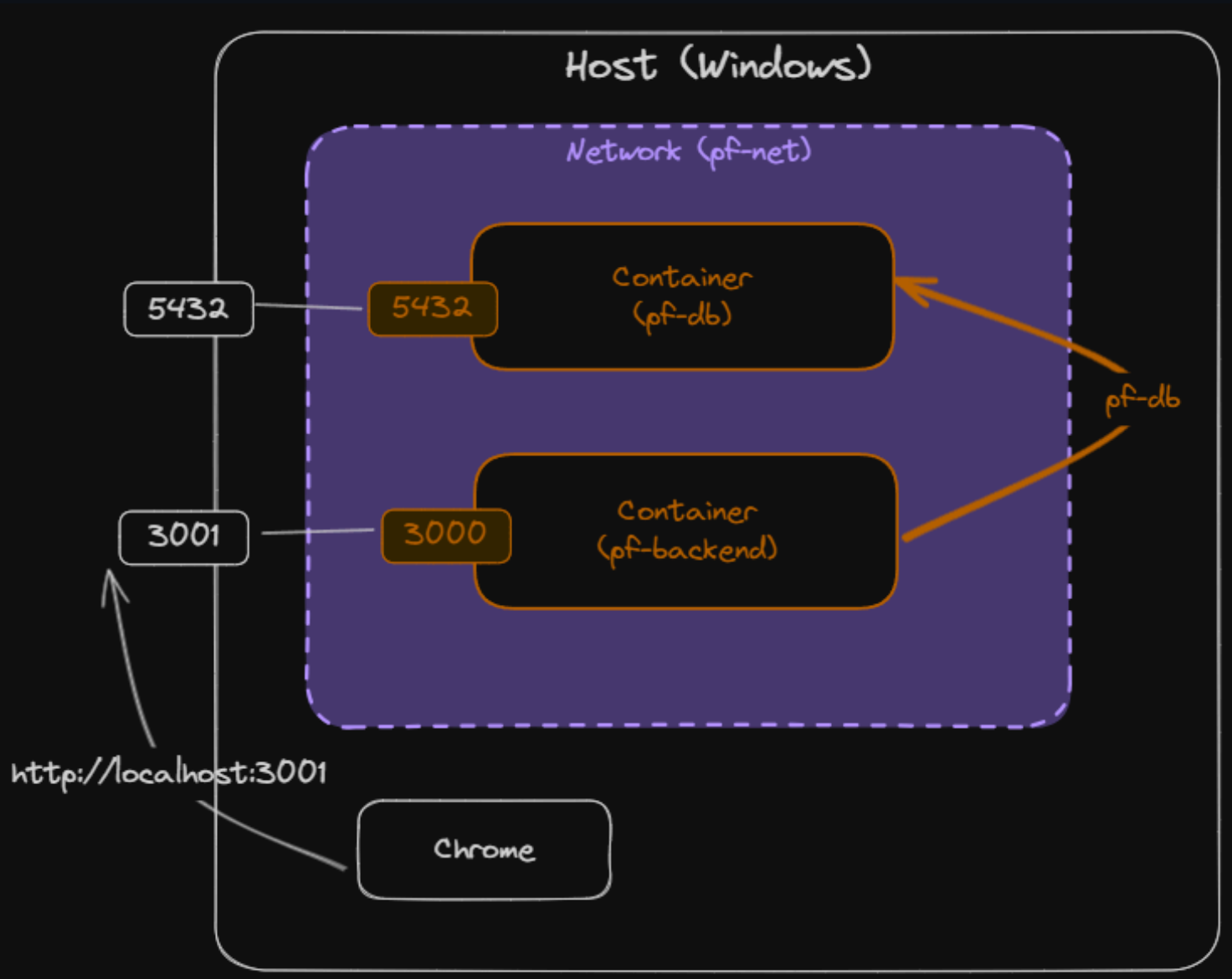
Dockerfile

- Double check the `pnpm` version in your local machine and the Dockerfile.



Network

- Notice the port `3001` / `3000`.
- *Can you find where these ports are specified?*



Automated database migration

- We want to make sure that database migrations are run automatically every time the container starts.
- We use the `post_start` hook in `docker-compose.yml`

Dockerhub

- Create an account at https://hub.docker.com.
- Create repository called `preflight-backend`.
- Login to your account in terminal
  - `docker login`

Push to Dockerhub

- Tag image
  - ⌨️ `docker tag preflight-backend [DOCKERHUB_ACCOUNT]/preflight-backend:latest`
- `docker login`
- Push image
  - ⌨️ `docker push [DOCKERHUB_ACCOUNT]/preflight-backend:latest`

Userful docker commands

- Inspect
  - `docker ps`
  - `docker network ls`
  - `docker volume ls`
- Cleaning
  - `docker image prune -a`
  - `docker builder prune`
  - `docker volume prune`
  - `docker network prune`